226186: sandstone, Nicolay 1

(Nambeet Formation, Canning Basin)

Location and sampling

Joanna Spring (SF 51-3), McTavish (3556)
MGA Zone 51, 526953E 7825419N

Sampled on 18 February 2019

This sample was collected from the 3496.0 – 3496.7 m depth interval of drillcore Nicolay 1, from a petroleum well drilled in 2012 by New Standard Onshore Pty Ltd (New Standard Onshore, 2013). The drillhole is located on a north–south cleared line, about 52 km south of Battlement Rocks, 30 km southeast of Turkey Place Hill, and 18.8 km west-northwest of McTavish Claypan.

Geological context

The unit sampled is the Early Ordovician (485–470 Ma; Cohen et al. 2020) Nambeet Formation of the Canning Basin (Haines, 2004). The unit is about 337 m thick in Nicolay 1 and consists mainly of shale with interbedded limestone and sandstone (New Standard Onshore, 2013). The Nambeet Formation is the lowermost Canning Basin unit in Nicolay 1 and is conformably overlain by the Willara Formation (Haines, 2004; Zhan, 2019). Three additional samples of Nambeet Formation sandstone from Nicolay 1 yielded maximum depositional ages of 589–533 Ma (GSWA 226183, Wingate et al., 2021a; GSWA 226184, Wingate et al., 2021b; GSWA 226185, Wingate et al., 2021c). A Nambeet Formation sandstone sample from Samphire Marsh 1, drilled about 246 km to the west-northwest, yielded detrital zircon ages of 2838–506 Ma, and a conservative maximum depositional age of 519 ± 3 Ma (GSWA 135962, Wingate and Haines, 2009). Another Nambeet Formation sample from Barnicarndy 1, drilled about 185 km to the southwest, yielded detrital zircon ages of 3274–590 Ma, and a conservative maximum depositional age of 1007 ± 18 Ma (GSWA 237998, Wingate et al., 2021d).

Petrographic description

The sample is a fine-grained calcareous sandstone, within which siltstone forms thin (<1.5 mm), wavy and lenticular to anastomosing laminae. The sample includes 15–20% carbonate as recrystallized sparry calcite, which is unevenly distributed within the fine-grained sandstone, and therefore probably cement. Carbonate-poor sandstone layers contain >50% quartz, <5% feldspar, and minor muscovite, pyrite, glauconite, collophane, and zircon. Quartz is angular and sub-angular, clear, and up to 0.15 mm in size. Feldspar grains are angular and up to 0.15 mm in size, and muscovite forms narrow laths up to 0.2 mm long. Green glauconite is up to 0.15 mm in size. Carbonate within these layers occurs as isolated rhombs and in patches of probable cement. Rare collophane rod-like features are probably fossil fragments. Carbonate-rich sandstone layers contain less quartz (~30%) and feldspar and more carbonate. Much of the carbonate that encloses the detrital grains is featureless spar, although there is a small amount of grains up to 0.3 mm in size, which have bioclastic textures and rare rods of carbonate or collophane that are probably fossil fragments. The silty laminae contain 10–25% detrital quartz grains with the remainder consisting of aligned white mica flakes, detrital muscovite laths up to 0.25 mm long, and brown clay. The siltstone layers lack carbonate. The sample contains circular framboidal pyrite grains, up to 0.025 mm, mainly within the siltstone layers. The sandstone has a mixed terrigenous and carbonate provenance with carbonate mainly occurring in the sandstone layers that also contain conspicuous fossil fragments. The more quartz-rich sandstone layers are well sorted although the grains are mainly angular and subangular and these layers, which lack obvious matrix, are texturally mature. The presence of fossil fragments and glauconite indicate a marine depositional environment and lenticular bedding is consistent with a relatively low-energy, tidal marine setting. The sample has not been deformed or metamorphosed.

Zircon morphology

Zircons isolated from this sample are mostly colourless to pale brown, anhedral to subhedral, and mainly strongly rounded. The crystals are up to 200 µm long, and equant to elongate, with aspect ratios up to 4:1. Some crystals have pitted outer surfaces and, in cathodoluminescence (CL) images, concentric zoning truncated at grain boundaries, features consistent with sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 11 June 2020, using LA-ICP-MS instruments at the GeoHistory Facility in the John de Laeter Centre, Curtin University. Separated zircons mounted in epoxy disks were ablated using a Resonetics RESOlution M-50A-LR sampling system, incorporating a 193 nm Compex 102 excimer laser, and ion signal intensities were measured using an Agilent 8900s quadrupole ICP-MS. Samples were ablated for 22 s at a 5 Hz repetition rate, using a 24 µm beam diameter and laser energy of 2.0 J/cm^2.

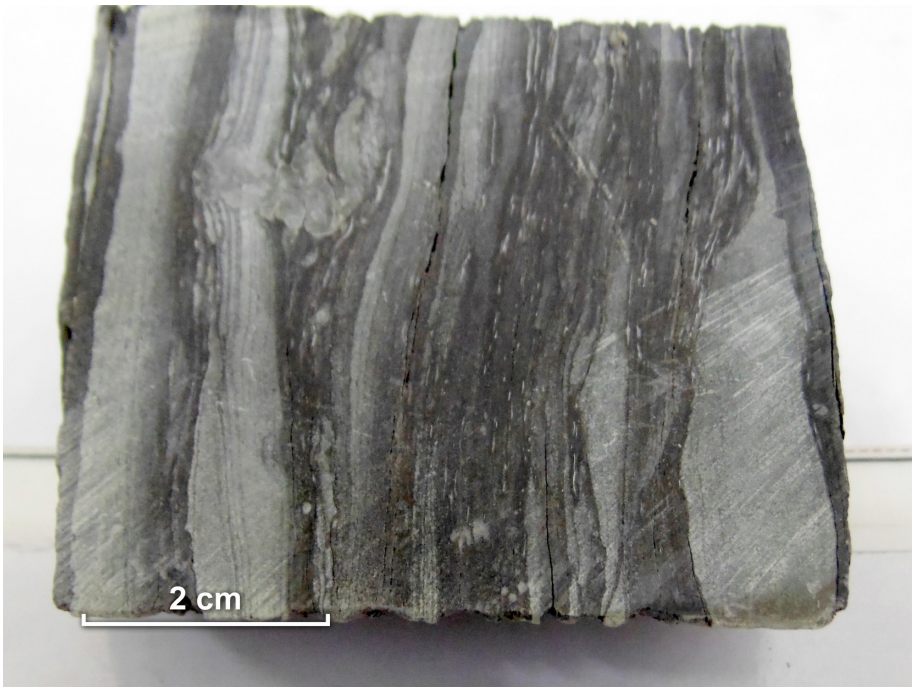

Figure 1. Drillcore image of sample 226186: sandstone, Nicolay 1

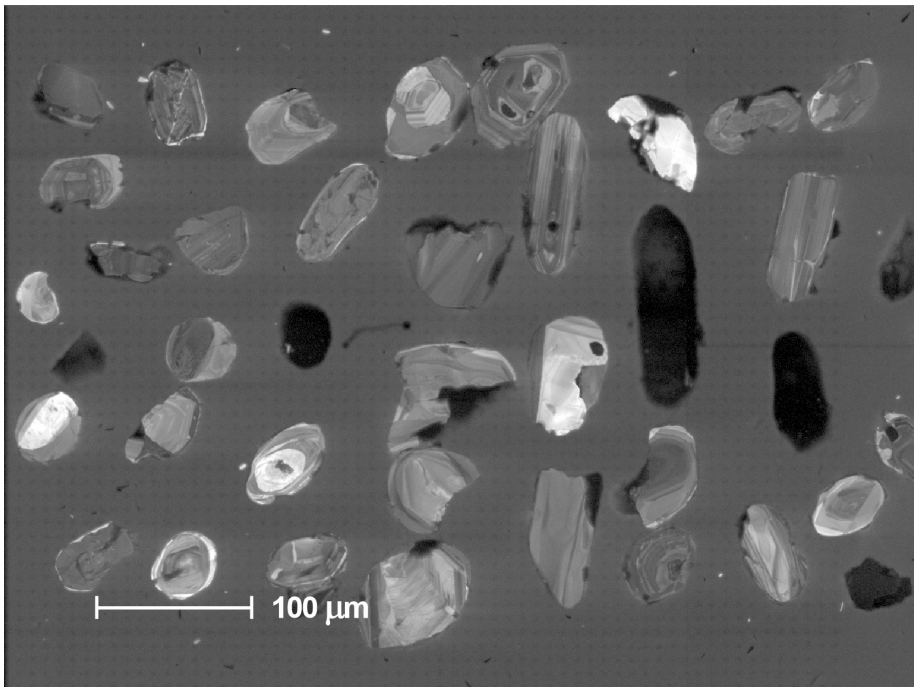

Figure 2. Cathodoluminescence image of representative zircons from sample 226186: sandstone, Nicolay 1

Table 1. LA-ICP-MS analytical results for zircons from sample 226186: sandstone, Nicolay 1

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$\frac{^{232}Th}{^{238}U}$	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		Disc. (%)
Y	4	4	307	97	0.32	11.669	0.082	0.06170	0.00110	530	4	664	38	20.1
S	12	12	651	267	0.41	10.084	0.056	0.06060	0.00090	610	3	625	32	2.5
S	2	2	711	123	0.17	7.559	0.131	0.06200	0.00305	801	13	674	105	-18.8
S	23	23	124	86	0.70	7.278	0.053	0.07160	0.00165	830	6	975	47	14.8
S	19	19	388	227	0.59	7.220	0.039	0.07100	0.00105	836	4	957	30	12.7
S	18	18	210	89	0.42	6.859	0.049	0.06820	0.00120	877	6	875	36	-0.3
S	17	17	274	106	0.39	6.523	0.047	0.07900	0.00120	919	6	1172	30	21.6
S	8	8	606	105	0.17	6.046	0.029	0.07210	0.00095	987	4	989	27	0.2
S	1	1	364	52	0.14	5.999	0.034	0.07750	0.00110	994	5	1134	28	12.4
S	24	24	932	95	0.10	5.718	0.031	0.07542	0.00090	1039	5	1080	24	3.8
S	20	20	200	56	0.28	5.255	0.032	0.07630	0.00120	1123	6	1103	31	-1.8
S	11	11	282	130	0.46	5.216	0.030	0.07660	0.00110	1131	6	1111	29	-1.8
S	13	13	446	691	1.55	4.819	0.023	0.07940	0.00105	1215	5	1182	26	-2.8
S	9	9	271	130	0.48	4.252	0.023	0.09020	0.00120	1362	7	1430	25	4.8
S	15	15	370	71	0.19	3.730	0.019	0.09350	0.00115	1531	7	1498	23	-2.2
S	10	10	368	67	0.18	3.799	0.020	0.09690	0.00120	1506	7	1565	23	3.8
S	16	16	133	74	0.56	3.470	0.020	0.09730	0.00140	1633	9	1573	27	-3.8
S	7	7	1184	216	0.18	2.946	0.013	0.11910	0.00135	1884	7	1943	20	3.0
D	3	3	1043	1123	1.08	6.061	0.119	0.08903	0.00105	985	18	1405	23	29.9
D	22	22	726	126	0.17	5.942	0.051	0.07610	0.00100	1003	8	1098	26	8.7
D	14	14	222	120	0.54	4.988	0.029	0.07660	0.00120	1178	6	1111	31	-6.1
D	5	5	154	133	0.87	5.313	0.044	0.07900	0.00130	1112	8	1172	33	5.2
D	21	21	582	553	0.95	3.545	0.018	0.10890	0.00130	1602	7	1781	22	10.1
D	6	6	509	203	0.40	3.209	0.020	0.11570	0.00135	1749	9	1891	21	7.5

Zircon standard 91500 was used as the primary reference material, and GJ1, OGC1 and Plešovice were included as secondary standards. Additional analytical details are provided in Fielding et al. (2021). All relevant uncertainties have been propagated into the ratios and dates reported in Table 1. Ages from analyses for which $^{238}U/^{206}Pb$ ratios indicate dates <1000 Ma are based on 207-corrected $^{238}U/^{206}Pb^*$ ratios; those >1000 Ma are based on uncorrected $^{207}Pb/^{206}Pb$ ratios.

Results

This sample yielded relatively few zircons, and 24 analyses were obtained from 24 zircons. Results are listed in Table 1, and shown in a concordia diagrams (Fig. 3) and probability density plots (Figs 4, 5).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Five analyses >1000 Ma are >5% discordant, and one analysis <1000 Ma indicates high common Pb (f_{207} >2%). The dates obtained from these six analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 18 analyses can be divided into two groups, based on their $^{207}Pb/^{206}Pb$ and $^{238}U/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a 207-corrected $^{238}U/^{206}Pb^*$ date of 528 ± 4 Ma (1σ).

Group S comprises 17 analyses (Table 1), which yield $^{207}Pb/^{206}Pb$ or 207-corrected $^{238}U/^{206}Pb^*$ dates of 1943–609 Ma.

It is possible that all of the analyses are of unmodified detrital zircons, in which case the date of 528 ± 4 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the sandstone. A more conservative estimate of the maximum depositional age can be based on the weighted mean 207-corrected $^{238}U/^{206}Pb^*$ date of 828 ± 7 Ma (MSWD = 2.0) for the youngest coherent group of three analyses in Group S.

The data for combined Groups Y and S indicate a significant age component at c. 1093 Ma, based on contributions from approximately 3 analyses (Fig. 5). This is interpreted as the age of the dominant zircon-crystallizing rock in the detrital source region, or the age of a dominant detrital component within sediments that have been reworked into this rock.

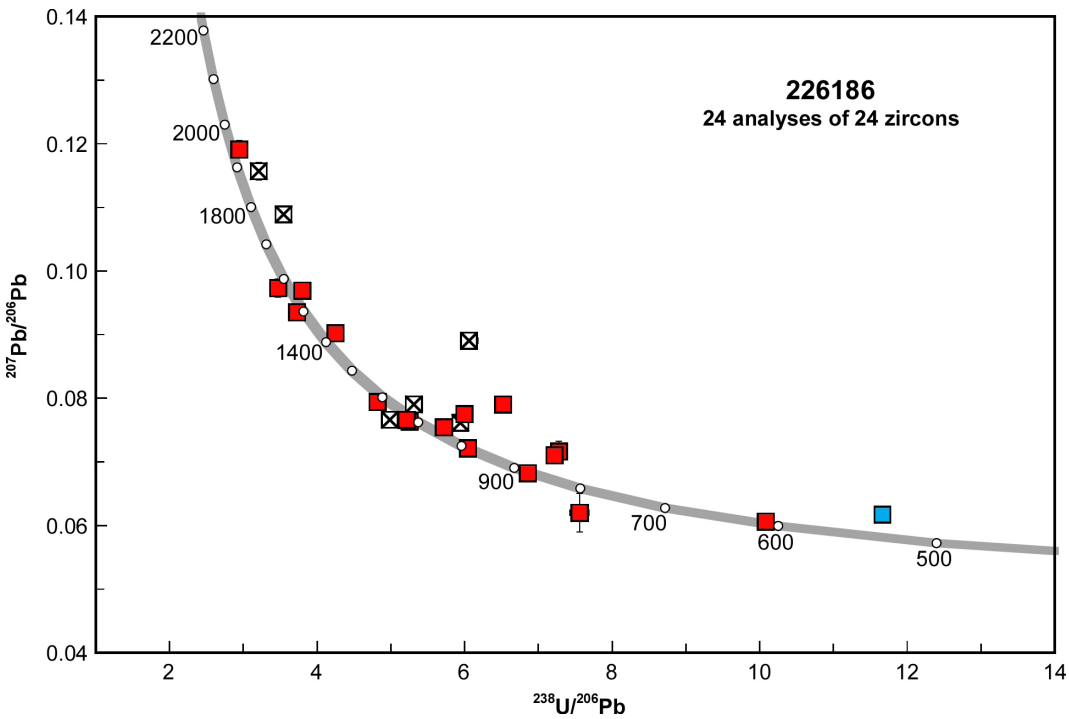

Figure 3. U–Pb analytical data, not corrected for common Pb, for zircons from sample 226186: sandstone, Nicolay 1. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (date >1000 Ma and discordance >5% or date <1000 Ma and f_{207} >2%)

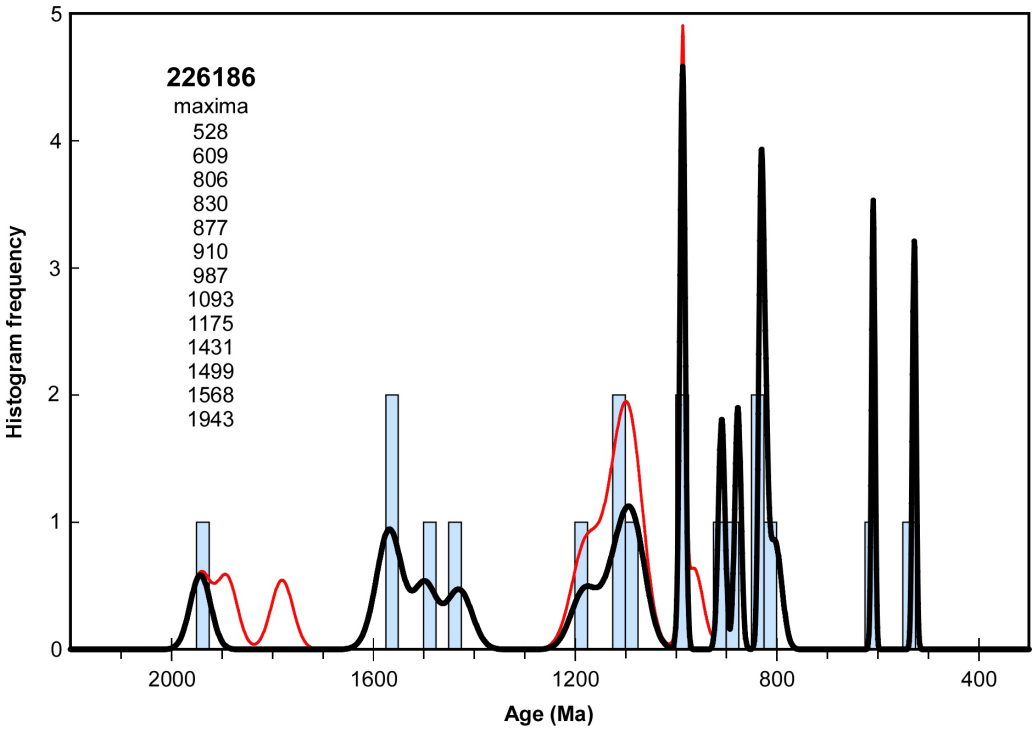

Figure 4. Probability density diagram and histogram for sample 226186: sandstone, Nicolay 1. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) only include accepted data (18 analyses of 18 zircons). Thin curve includes all data (24 analyses of 24 zircons)

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Recommended reference for this publication

Wingate, MTD, Lu, Y and Dent, LM 2021, 226186: sandstone, Nicolay 1; Geochronology Record 1763: Geological Survey of Western Australia, 5p.

Data obtained: 11 June 2020

Data released: 14 May 2021